# L11: Kuramoto Model on a Complete Network

The Kuramoto model has been studied in several different variations:

A. Is the number of oscillators *N* finite or can we assume that *N tends to infinity*? *(the former is harder to model)*

B. Do the oscillators have the same natural frequency or do their frequencies follow a given statistical distribution? *(the latter is harder to model)*

C. Is each oscillator coupled with every other oscillator or do these interactions take place on a given network? *(the latter is harder to model)*

D. Is the coupling between oscillators instantaneous or are there coupling delays between oscillators? *(the latter is **much** harder to model)*

In the next couple of pages we will review some key results for the asymptotic case of very large N, and without any coupling delays.

Let us start with the simpler case in which each oscillator is coupled with every other oscillator:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\phi_j - \phi_i)$$

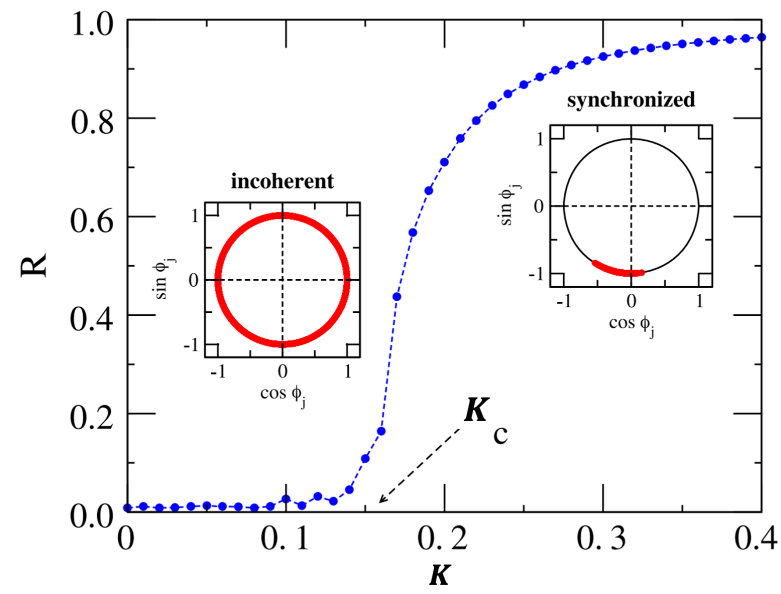
The initial phase of each oscillator is randomly distributed in $[0, 2\pi)$. The coupling network between oscillators, in this case, can be thought of as a clique with equally weighted edges: each oscillator has the same coupling strength with every other oscillator.

Further, let us assume that the natural angular velocities of the *N* oscillators follow a unimodal and symmetric distribution around its mean $E[\omega_i] = \Omega$.

When do these *N* oscillators get synchronized? It all depends on how strong the coupling strength *K* is relative to the maximum difference of the oscillator natural frequencies.

If *K* is close to *0* the oscillators will move at their natural frequencies, while if *K* is large enough, we would expect the oscillators to synchronize.

Intuitively, we may also expect an intermediate state in which *K* is large enough to keep smaller groups of oscillators synchronized – but not all of them.




The visualization shows the asymptotic value of the order parameter *R(t) (as t tends to infinity)*, as a function of the coupling strength *K*.

For smaller values of *K*, the oscillators remain incoherent *(i.e., not synchronized)*.

As *K* approaches a critical value $K_c$ *(around 0.16 in this case)*, we observe a phase transition that is referred to as the ***"onset of synchronization".***

For larger values of $K > K_c$, the order parameter *R* increases asymptotically towards one, suggesting that the oscillators get completely synchronized.

For the asymptotic case of very large *N*, Kuramoto showed that as *K* increases, the system of *N* oscillators shows a phase transition at a critical threshold $K_c$.

At that threshold, the system moves rapidly from a de-coherent state to a coherent state in which almost all oscillators are synchronized *(and so the order parameter R increases rapidly from 0 to 1).*

A necessary condition for the onset of complete synchronization is that

$K \geq K_c = \frac{2}{\pi\Omega}$

This visualization shows numerical results from simulations with *N=100* oscillators in which the average angular frequency is $\Omega = 4$ rads/sec, and $K_c \approx 0.16$.

## Food For Thought

Try to derive the Kuramoto critical coupling threshold given on this page.